RFC 5681: TCP Congestion Control Specification

Abstract:
This document defines four intertwined TCP congestion control algorithms:
slow start, congestion avoidance, fast retransmit, and fast recovery.

Section 1: The Four Algorithms
1. Slow Start:
During slow start, TCP increments cwnd by at most SMSS bytes for each ACK received that acknowledges new
Slow start ends when cwnd exceeds ssthresh or when congestion is observed.

2. Congestion Avoidance:
Once cwnd is greater than ssthresh, cwnd is increased by approximately 1 SMSS per round trip time (RTT).

3. Fast Retransmit:
Upon receipt of three duplicate ACKs, TCP immediately retransmits the missing segment without waiting for the

4. Fast Recovery:
Governs the transmission of new data until a non-duplicate ACK arrives, avoiding an unnecessary drop to cwnd

Section 2: Terminology and Constants
- SMSS (Sender Maximum Segment Size): The size of the largest segment that the sender can transmit.
- IW (Initial Window): Typically set between 2 and 4 segments, modernized to 10 SMSS.
- FlightSize: The amount of data that has been sent but not yet acknowledged.